The Hecke Algebra Was Already There

what `q-bool` and `lascoux` contain of the Langlands programme, and the one chapter that closes the gap

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Abstract

Two repositories built for unrelated reasons — `q-bool`, on containers over a finite field, and `lascoux`, on the plactic monoid as a container — turn out to contain, unnamed, several of the standard objects of the Langlands programme. The central observation is not an analogy: the operator S_i of `schubert.pdf`, defined there as pushforward along a forgetful map of flags followed by pullback, *is* the Iwahori–Hecke generator, and the relations verified there are a presentation of $H_n(q)$. We re-derive this from the geometry independently of `flag.py` (§3); identify the refuted nilCoxeter conjecture's true target as the Grothendieck polynomials of Lascoux–Schützenberger (§4); locate the Langlands dual group in the Bernstein centre of the *affine* Hecke algebra, one level above where `q-bool` currently works (§5); verify that the Hall constants of `q-container.pdf` §9 are the Hall–Littlewood constants, which by Macdonald's Chapter V are the spherical Hecke algebra of GL_n over a local field (§6); identify the two-parameter ceiling of `q-container.pdf` with the loop direction, i.e. with Cherednik's double affine Hecke algebra (§7); and show that `q-dirichlet.pdf` §7 has already reached the automorphic side of function-field Langlands (§8). For `lascoux` we record three links, one of which — $f^{\lambda}_{\mu\nu}(0) = c^{\lambda}_{\mu\nu}$ — is a bridge between the two repositories rather than to Langlands, and is verified here. Four computations, all independent of the theorems they check, are in `code/verify.py`. §10 records what does *not* link, and §12 the status of every claim.

1 The question

`q-bool` was written to ask what containers become over $\mathbb{F}_q$. `lascoux` was written to ask whether Fomin's local rules are a container isomorphism. Neither was written with the Langlands programme in view, and neither mentions it.

The mishnah is therefore: nothing should connect. Container theory is a fragment of dependent type theory; Langlands is arithmetic. The shared vocabulary — q, flags, Hall, Macdonald — is the vocabulary of algebraic combinatorics, which both subjects borrow from, and borrowing a word is not sharing an object.

The objection is Proposition 4.1 of `schubert.pdf`. It records, as a computational finding about an operator built from a fibration of flag varieties, three relations:

$$T_i^2 = (q-1)T_i + qI, \qquad T_iT_jT_i = T_jT_iT_j \ (|i-j|=1), \qquad T_iT_j = T_jT_i \ (|i-j|\geq 2).$$

That is not a family resemblance. It is a presentation, and the algebra it presents has a name.

2 Hecke operators, in one page

For orientation, the object in its five standard guises. Only the first three are needed below.

Sublattices. A modular form of weight k is a function on lattices $\Lambda \subset \mathbb{C}$, homogeneous of degree $-k$. For a prime p there are $p+1$ sublattices of index p, and no canonical choice among them, so one sums:

$$(T_p F)(\Lambda) \;=\; \sum_{[\Lambda:\Lambda']=p} F(\Lambda').$$

On q-expansions, $T_p f = \sum_n (a_{np} + p^{k-1} a_{n/p}) q^n$; for an eigenform normalised with $a_1 = 1$, the Fourier coefficients *are* the eigenvalues, which is why $L(s, f)$ has an Euler product.

Double cosets. T_p is the double coset $[\Gamma \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma]$. Adelically, for a local field F with ring of integers $\mathcal{O}$ and residue field $\mathbb{F}_q$, the *spherical* Hecke algebra is $C_c(K \backslash G / K)$ under convolution with $K = G(\mathcal{O})$; the *Iwahori* Hecke algebra takes instead $K =$ the preimage of a Borel. The two differ, and the difference is exactly the gap identified in §5.

Satake. The spherical Hecke algebra is commutative, and

$$\mathcal{H}\big(G(F), G(\mathcal{O})\big) \;\cong\; \mathbb{C}[X_*(T)]^W \;\cong\; \mathrm{Rep}\big({}^L G\big),$$

the representation ring of the Langlands dual group. A Hecke eigenvalue is thus not a number but a semisimple conjugacy class in ${}^L G(\mathbb{C})$.

Trees. For $G = \mathrm{GL}_2$, the quotient $G(F)/G(\mathcal{O})F^\times$ is the vertex set of the $(q+1)$-regular Bruhat–Tits tree, and the Hecke operator is its adjacency operator. Deligne's bound $|a_p| \le 2p^{(k-1)/2}$ is the optimal spectral gap, which is where Ramanujan graphs come from.

Correspondences. Geometrically, a Hecke operator modifies a G-bundle at one point: the stack of (E, E', x, φ) with φ an isomorphism away from x, with its two projections to Bun_G. The type of modification is a cocharacter of G, i.e. a weight of ${}^L G$ — Satake again.

Each guise is the same refusal: an operation defined only up to a finite set of choices at one place, and a sum instead of a choice.

3 `schubert.pdf` constructs the Iwahori–Hecke algebra

Definition 3.1. Let $\mathrm{Fl}_n(\mathbb{F}_q)$ be the set of complete flags in $\mathbb{F}_q^n$, π_i the map forgetting the i-th subspace, and

$$(S_i f)(F) \;=\; \sum_{\pi_i(F') = \pi_i(F)} f(F').$$

This is Definition 3.1 of `schubert.pdf`, verbatim.

Theorem 3.2. *S_i is convolution by the indicator function of the i-th minimal parabolic P_i, hence $T_i := S_i - I$ is convolution by $\mathbf{1}_{Bs_iB}$; the T_i generate the Iwahori–Hecke algebra $H_n(q)$ of type A_{n-1}, and*

$$\mathrm{End}_{\mathrm{GL}_n(\mathbb{F}_q)}\big(\mathrm{Ind}_B^G \mathbf{1}\big) \;=\; \mathrm{End}_{\mathrm{GL}_n(\mathbb{F}_q)}\big(\mathbb{C}[G/B]\big) \;\cong\; H_n(q).$$

Proof. Functions on $\mathrm{Fl}_n(\mathbb{F}_q) = G/B$ with G acting; π_i is the projection $G/B \to G/P_i$, so pushforward-then-pullback is convolution by $\mathbf{1}_{P_i}$. The Bruhat decomposition of a minimal parabolic is $P_i = B \sqcup Bs_iB$, so $\mathbf{1}_{P_i} = \mathbf{1}_B + \mathbf{1}_{Bs_iB}$ and $T_i = S_i - I$ is convolution by $\mathbf{1}_{Bs_iB}$. Since $|Bs_iB/B| = q$, the fibre of π_i has $q+1 = |\mathbb{P}^1(\mathbb{F}_q)|$ points and $\mathbf{1}_{Bs_iB} * \mathbf{1}_{Bs_iB} = (q-1)\mathbf{1}_{Bs_iB} + q\,\mathbf{1}_B$. The braid relations are the braid relations in $W = S_n$ lifted through the Bruhat decomposition. The last display is Iwahori's theorem [1]. □

Remark 3.3. This is what Proposition 4.1 of `schubert.pdf` records, one level of abstraction down: there the relations are found by computing the matrix of S_i on cell classes and checking. The identification above says why they were always going to hold, and that the cell rule quoted in Example 3.3 there,

$$S_i[w] = \begin{cases} [w] + [ws_i], & \ell(ws_i) > \ell(w), \\ q[w] + q[ws_i], & \ell(ws_i) < \ell(w), \end{cases}$$

is the standard $H_n(q)$ action on the Bruhat basis.

Proposition 3.4. ***verified.*** *`code/verify.py` rebuilds* $\mathrm{Fl}_n(\mathbb{F}_q)$ *from subspace enumeration, computes the relative position from the rank matrix of Definition 3.1, checks* $|\mathrm{Fl}_n| = [n]_q!$ *and* $|C_w| = q^{\ell(w)}$*, checks every fibre of* π_i *has* $q+1$ *elements, checks that* S_i *applied to a cell indicator is constant on cells, and then verifies* $S_i^2 = (q+1)S_i$*, the Hecke quadratic, both braid relations and* $T_i^2 \neq 0$*, for* $(n,q) \in \{(2,2),(2,3),(3,2),(3,3),(4,2)\}$ *— all* 315 *flags and* 24 *cells in the largest case. It shares no code with `q-bool/code/flag.py` and does not assume the cell rule.*

4 The refutation relocated: Grothendieck, not Schubert

`schubert.pdf` proves that no nilpotent divided difference exists on the flag container over any finite field, and concludes that the nilCoxeter half of its Conjecture is refuted. Both are correct. What the document does not say is that the object it *did* find has a Schubert-like basis of its own.

Proposition 4.1. *The* $q \to 0$ *degeneration of* $H_n(q)$ *is the* 0*-Hecke algebra, generated by idempotents* π_i *with* $\pi_i^2 = \pi_i$ *— Proposition 4.1(a) of `schubert.pdf` is this statement for* $S_i/(q+1)$*. The* π_i *are the Demazure operators; they act on* $K(G/B)$*; and the basis they generate is the Grothendieck polynomials* $\mathfrak{G}_w$ *of Lascoux–Schützenberger [2].*

So the flag container does not fail to produce a Schubert-type basis. It produces the K-theoretic one. This is forced: counting $\mathbb{F}_q$-points is a K-theoretic invariant, and Schubert polynomials $\mathfrak{S}_w$ live in the Chow ring. The table in §5 of `schubert.pdf` states exactly this dichotomy; what is added here is that the left-hand column has a named polynomial basis, and that the name is Lascoux — the subject of the sibling repository.

Proposition 4.2. *`schubert.pdf` Question 5.2 (grade the positions of the flag container by the torus action; do the divided differences appear?) is answered by the nilHecke ring of Kostant–Kumar [3], acting on* $H_T^*(G/B)$*, in which* ∂_i *and* $\mathfrak{S}_w$ *live; Lusztig's graded affine Hecke algebra [4] is the object interpolating between it and* $H_n(q)$*. The question is not open.*

Remark 4.3. The failure point $q = -1$ of `schubert.pdf` Theorem 4.2 is the arithmetic signature of this dichotomy rather than an accident: $q = -1$ is where a point count degenerates to an Euler characteristic.

5 Where the dual group is, and why `q-bool` cannot yet see it

Theorem 3.2 gives the Iwahori-level algebra over a *finite* field. The Langlands dual group is not visible there.

Proposition 5.1. *The centre of the finite Hecke algebra* $H_n(q)$ *is the algebra of symmetric polynomials in the Jucys–Murphy elements — a shadow of* $\mathrm{Rep}(\mathrm{GL}_n)$*, but finite-dimensional and carrying no lattice. By contrast, for the affine Hecke algebra* $\widetilde{H}$ *of* $G(F)$*, Bernstein's theorem gives*

$$Z(\widetilde{H}) \;\cong\; \mathbb{C}[X_*(T)]^W \;\cong\; \mathrm{Rep}({}^L G).$$

The Langlands dual group is the centre of the affine algebra, not of the finite one.

Remark 5.2. `q-bool` is already at $\mathcal{O}$ in one document and at $\mathbb{F}_q$ in another, and has not joined them. `q-container.pdf` §9 passes from $\mathbf{Vect}_{\mathbb{F}_q}$ to finite modules over a discrete valuation ring with residue field $\mathbb{F}_q$ — the lattice side. `schubert.pdf` works with flags over $\mathbb{F}_q$ — the flag side. The same local field underlies both. What is missing is one construction.

Question 5.3. **open**. Define the *affine flag container*: shapes the affine Weyl group $\widetilde{W} = \mathbb{Z}^n \rtimes S_n$, positions $\mathbb{F}_q^{\ell(w)}$, obtained by replacing complete flags in $\mathbb{F}_q^n$ by complete lattice chains in $\mathcal{O}^n$. Three things should then be checked, in this order.

(i) The counting series $\sum_{w\in\widetilde{W}} X^{\ell(w)}$ is no longer a polynomial. Bott's formula predicts a rational function; confirming which one is the cheapest possible test that the construction is right.

(ii) The operators S_i, now indexed by $i = 0, 1, \dots, n-1$, satisfy the affine braid relations together with the same quadratic.

(iii) The Bernstein centre of the resulting algebra is $\mathbb{C}[x_1^{\pm}, \dots, x_n^{\pm}]^{S_n} = \mathrm{Rep}(\mathrm{GL}_n(\mathbb{C}))$.

Step (iii), if it goes through, is the Langlands dual group appearing inside a container.

The GL_2 case of the relevant spherical identity is verified in `verify.py`, by enumerating sublattice chains in $\mathbb{Z}_p^2$ rather than by quoting the Hecke ring:

Proposition 5.4. ***verified***. *For $p \in \{2, 3, 5\}$: there are $p+1$ sublattices of index p and p^2+p+1 of index p^2, of which exactly one has elementary divisors (p, p) and p^2+p have $(1, p^2)$; the number of chains $\mathbb{Z}^2 \supset \Lambda' \supset \Lambda''$ with both steps of index p is $(p+1)^2$, distributed as multiplicity $p+1$ over the (p, p) lattice and multiplicity 1 over each $(1, p^2)$ lattice. Hence*

$$T_{(p,1)}^2 \;=\; T_{(p^2,1)} \;+\; (q+1)\, T_{(p,p)},$$

whose normalised Satake image is $(x+y)^2 = (x^2+y^2) + 2xy$ in $\mathbb{Z}[x+y, (xy)^{\pm 1}] = \mathrm{Rep}(\mathrm{GL}_2)$.

6 Hall–Littlewood is the spherical Hecke basis

This is the strongest link, and the cheapest to act on.

Proposition 6.1. ***verified***. *Let $g_{\mu\nu}^{\lambda}(p)$ count submodules $U \subseteq M_\lambda$ over $\mathcal{O}$ with $U \cong M_\nu$ and $M_\lambda/U \cong M_\mu$, and let $f_{\mu\nu}^{\lambda}(t)$ be defined by $P_\mu(x;t)P_\nu(x;t) = \sum_\lambda f_{\mu\nu}^{\lambda}(t) P_\lambda(x;t)$. Then*

$$g_{\mu\nu}^{\lambda}(p) \;=\; p^{\,n(\lambda)-n(\mu)-n(\nu)}\, f_{\mu\nu}^{\lambda}(p^{-1}).$$

Verified over 128 instances, all λ with $|\lambda| \le 4$, $p \in \{2, 3\}$: submodules enumerated by brute-force closure in $\mathbb{Z}/p^N$; Hall–Littlewood polynomials computed symbolically from the Weyl-type formula in four variables; neither side quoted from Macdonald.

That is Hall's theorem, and it is the content of `q-container.pdf` §9, which is stated there as classical but unverified. What §9 does not say is what Chapter V of the same book of Macdonald's says:

Theorem 6.2 (Macdonald [5], Ch. V)**.** *The spherical Hecke algebra $\mathcal{H}(\mathrm{GL}_n(F), \mathrm{GL}_n(\mathcal{O}))$ of a local field with residue field $\mathbb{F}_q$ is isomorphic to the Hall algebra of finite $\mathcal{O}$-modules; its zonal spherical functions are the Hall–Littlewood polynomials $P_\lambda(x; q^{-1})$; and composing with Satake gives*

$$\textit{Hall algebra of finite } \mathcal{O}\textit{-modules} \;\cong\; \mathcal{H}\big(\mathrm{GL}_n(F), \mathrm{GL}_n(\mathcal{O})\big) \;\cong\; \mathrm{Rep}\big({}^L\mathrm{GL}_n\big).$$

Remark 6.3. `q-container.pdf` §9 already writes the normalisation $P_\lambda(x; q^{-1})$ — Macdonald's Chapter V normalisation — and already computes the Euler-form twist $q^{\langle M,N\rangle}$ relating the Hall algebra to its twisted form. The book has one title because its two halves are one subject, and the hinge is Chapter V. Reading it against §9 is the single highest-yield next step in this document: it decides whether `q-bool` contains a Langlands dual group or stops at Hall–Littlewood.

7 The two-parameter ceiling is the loop direction

`q-container.pdf` §*Ceiling* observes that Macdonald's $P_\lambda(x; q, t)$ has two parameters, that neither is the field size, that the theory as built lands at $q_{\text{Mac}} = 0$ (Hall–Littlewood), and that a second direction is needed, with three candidates and no way to choose.

Claim 7.1. The second direction is the loop. In Cherednik's double affine Hecke algebra [6], of which Macdonald polynomials are the eigenfunctions: t is the Hecke parameter — the field size, the $q + 1$ of Definition 3.1, the Iwahori level — and q is the parameter of the translation lattice, i.e. loop rotation. The specialisation $q_{\text{Mac}} \to 0$ giving Hall–Littlewood is exactly the specialisation that forgets the loop.

Remark 7.2. The loop variable is already present in `q-container.pdf`, written t, as the uniformiser of $\mathcal{O} = \mathbb{F}_q[[t]]$. If Claim 7.1 is right then the document's own two letters are Macdonald's two parameters, and the "second direction" is not a new construction but a grading by t-adic degree on a structure already defined. This is a prediction and is not verified here.

8 `q-dirichlet.pdf` §7 is the automorphic side

`q-dirichlet.pdf` §7 replaces the monoid of monic polynomials by the monoid of effective divisors on a smooth projective curve $X/\mathbb{F}_q$, obtains $Z_X(u)$, and closes with a remark that categorifying the fibres — replacing divisors by coherent sheaves — gives Kapranov's Hall algebra of X.

That remark is the door. Unramified automorphic forms for GL_n over the function field $\mathbb{F}_q(X)$ are functions on $\mathrm{Bun}_n(\mathbb{F}_q)$; the Hecke operators are modifications at closed points, which is §2 decategorified from D-modules to point counts; and Kapranov [7] makes the Hall algebra of $\mathrm{Coh}(X)$ act by precisely these operators, recovering Eisenstein series. This is the setting of Drinfeld's and Lafforgue's theorems.

Remark 8.1. The document's own observation that the Euler product is the statement that Day convolution respects free generation is the right categorical account of why automorphic L-functions factor over places; and the local factor at a closed point P is where the Satake parameter, a conjugacy class in ${}^L G$, lives.

9 `lascoux`: three links, one of them internal

(a) Crystals are the geometric-Satake shadow of the dual group. Geometric Satake [8] gives $\mathrm{Perv}_{G(\mathcal{O})}(\mathrm{Gr}_G) \cong \mathrm{Rep}({}^L G)$ as tensor categories, and the Mirković–Vilonen cycles carry a canonical crystal structure isomorphic to Kashiwara's [9, 10]. `lascoux-container.pdf` §"crystals are directed containers" is therefore one functor from $\mathrm{Rep}({}^L G)$. Note also that $\mathrm{Gr}_G = G(F)/G(\mathcal{O})$ is the space on which the spherical Hecke algebra of §6 acts: crystals and Hall–Littlewood are two views of one space.

(b) The plactic constants are the Hall constants at $t = 0$. This is a bridge between the two repositories that does not pass through Langlands at all.

Proposition 9.1. ***verified.*** *$f^{\lambda}_{\mu\nu}(0) = c^{\lambda}_{\mu\nu}$, the Littlewood–Richardson coefficient. Verified over 64 instances, $|\lambda| \leq 4$: the left side from the symbolic Hall–Littlewood expansion of Proposition 6.1, the right side by independent brute-force enumeration of Littlewood–Richardson skew tableaux (rows weakly increasing, columns strictly increasing, reverse reading word a lattice word), itself sanity-checked against eight known coefficients.*

`lascoux-container.pdf` asserts that the plactic Schur elements S_λ multiply by Littlewood–Richardson; `q-container.pdf` §9 computes multiplication by $g^{\lambda}_{\mu\nu}(q)$. Proposition 9.1 plus Proposition 6.1 say these are the $t = 0$ and generic-t faces of one structure. The plactic monoid is the degenerate fibre of the Hall algebra.

(c) The q-gap and the Macdonald ceiling are the same gap. The section of `lascoux-container.pdf` headed "The gap: q, LLT, and Hecke algebras" states that there is no candidate route from a polynomial functor to a quantum group. LLT concerns Hecke algebras at roots of unity, and Ariki's theorem [11] identifies their decomposition numbers with the canonical basis of $U_v(\widehat{\mathfrak{sl}}_e)$ — a quantum *affine* algebra.

Claim 9.2. The missing direction in `lascoux` (LLT, quantum affine) and the missing direction in `q-bool` (Macdonald's second parameter, Claim 7.1) are the same direction: the loop. One gap, two repositories, two names.

This is my claim, not a theorem. It is strongly suggested by the fact that both resolutions are affine objects over the same base, and it predicts that whichever repository first constructs the loop grading closes both gaps at once.

10 What does not link

Recorded so that the ledger is not only positive.

`finite-field-duality.pdf`.
The classification of nondegenerate forms over $\mathbb{F}_q$ is the classification of the groups on which the non-self-dual Langlands duality acts (${}^L\mathrm{SO}(2n+1) = \mathrm{Sp}(2n)$), so the input data coincide. But nothing in the document uses the root datum, and the coincidence is of subject matter, not of construction. Weak.

`error-correcting.pdf`, `data-control.pdf`.
No link found. The data/control distinction is about variance in a container morphism; the Hecke correspondence is not a container morphism in that sense, because both legs move shapes. If there is a link it is through the flag container's algebra of dispatch (`data-control.pdf` §4), which by Theorem 3.2 is $H_n(q)$ — but what that buys is not clear, and is not claimed here.

Arithmetic Langlands.
Nothing above reaches number fields. Everything is over $\mathbb{F}_q$ or a local/function field. The Galois side never appears.

11 *Mai nafka minah*

What changes in practice, in order of cost.

1. Read Macdonald Chapter V against `q-container.pdf` §9. Short chapter, input already in your notation, decides Theorem 6.2's applicability. Highest yield.

2. Amend `schubert.pdf`: name the algebra $H_n(q)$ (Theorem 3.2), retitle the refutation around Grothendieck polynomials (Proposition 4.1), close Question 5.2 with Kostant–Kumar (Proposition 4.2).
3. Add Proposition 9.1 to `lascoux`: the plactic constants are the Hall constants at $t = 0$. Cheap, verified, and it is the only statement joining the two repositories.
4. Build the affine flag container (Question 5.3), test (i) first. This is the only route to ${}^L G$ from where `q-bool` stands.
5. Decide Claim 7.1 by computing whether the t-adic grading on $\mathcal{O} = \mathbb{F}_q[[t]]$ induces Macdonald's q. If yes, Claim 9.2 follows and both gaps close.

12 Status of every claim

#	claim	where	status
1	S_i is convolution by $\mathbf{1}_{P_i}$; T_i generate $H_n(q)$	Thm. 3.2	**classical** (Iwahori); proof given
2	relations hold, re-derived from the geometry	Prop. 3.4	**verified** — 5/5 cases, 315 flags
3	$q \to 0$ gives Demazure, basis is $\mathfrak{G}_w$	Prop. 4.1	**classical** (Lascoux–Sch.)
4	Question 5.2 is Kostant–Kumar nilHecke	Prop. 4.2	**classical**; the question is not open
5	${}^L G$ is the Bernstein centre of the *affine* algebra	Prop. 5.1	**classical** (Bernstein)
6	GL_2 spherical identity and its Satake image	Prop. 5.4	**verified** — $p \in \{2, 3, 5\}$
7	Hall's theorem: $g^\lambda_{\mu\nu} \leftrightarrow$ Hall–Littlewood	Prop. 6.1	**verified** — 128 instances
8	Hall algebra $\cong$ spherical Hecke $\cong \mathrm{Rep}({}^L\mathrm{GL}_n)$	Thm. 6.2	**classical** (Macdonald Ch. V); unverified
9	Macdonald's second parameter is the loop	Cl. 7.1	**classical** (Cherednik); application **open**
10	`q-dirichlet` §7 is the GL_n automorphic side	§8	**classical** (Kapranov); door left closed
11	crystals $\leftrightarrow$ MV cycles $\leftrightarrow \mathrm{Rep}({}^L G)$	§9(a)	**classical** (Mirković–Vilonen)
12	$f^\lambda_{\mu\nu}(0) = c^\lambda_{\mu\nu}$: the repo bridge	Prop. 9.1	**verified** — 64 instances
13	the LLT gap and the Macdonald gap are one gap	Cl. 9.2	**open** — my claim, unproved
14	affine flag container has $\mathrm{Rep}({}^L\mathrm{GL}_n)$ as its centre	Q. 5.3	**open** — not attempted
15	`finite-field-duality` links to root data	§10	weak; subject matter only
16	anything reaches arithmetic Langlands	§10	no

13 Related work

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